Lock Before You Talk: Pooling First Reads Beats Deliberation When LLM Panels Split the Evidence

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Abstract

Safety designs often let a panel of AI agents discuss before it decides. We tested this with panels of six open-weight models from four families, solving 30 fictional whodunit cases with known answers, either alone, after two rounds of discussion, or with the evidence split between them. Two audits of our first analysis changed its results, and we report the corrected ones. First, our original agreement measure compared probability lists within each case, which scores six agents that each solve a case alone as 1.02 effective forecasters. Measured as error correlation across cases, discussion made errors line up in all four arms (free discussion: r 0.42 → 0.77; two of four intervals exclude zero; 5 to 14 model pairs). Second, with split evidence, the average of all locked first reads named the culprit in 60 of 60 panels and the discussed vote in 53 (exact McNemar p = 0.016). This result is suggestive, not established: on the same agents at both stages it is 57 against 53 (p = 0.29), and 19% of discussion statements were empty because of reasoning truncation. Third, our "same-model premium" came from one model's one-hot certainty, and we retract it. Recording and pooling independent first reads is cheap; whether it beats discussion needs the pre-registered follow-up we describe.

1. Introduction

When several AI agents check the same thing, a natural design lets them talk first and then vote. Talk can share evidence. It can also make agents copy each other, so that a panel of six carries the information of one. Which effect wins decides whether discussion makes multi-agent oversight safer.

We built a controlled test. Panels of six models judge fictional village cases with a known culprit, so every answer can be scored without an AI judge. We compare three decision rules on the same panels: the average of each agent's first read before any talk, the average after two rounds of discussion, and agents told that their first read is locked. We also split the evidence, so that each agent sees half of the clues.

This report differs from our first draft. Two same-weekend audits, run separately, found that our agreement measure counted correct, independent agents as one voice, that our "same-model premium" was an artifact, and that the harness never showed agents their locked first reads. We kept the pre-registration unchanged and report what holds up after the fixes.

Contributions.

- A simulation test showing that within-case agreement confuses being right with herding, and a re-measurement on the right quantity, error correlation across cases, which rises after discussion.
- Suggestive evidence that pooling locked first reads beats a discussed vote when evidence is split (60 of 60 against 53 of 60; 57 against 53 on matched agents), with the data-quality limits that keep it from being conclusive.
- A retraction or revision of four earlier claims, with the logs and two audits to check them.

2. Related Work

Correlated errors in AI crowds. Kim et al. (2025) find that across more than 350 LLMs, models make strongly correlated mistakes. Spiro (2026) reports error correlations of 0.74 to 0.82 between frontier forecasters, and Begin et al. (2026) find that preference tuning lowers ten forecasters to about 1.4 effective ones. These studies measure static forecasts. We measure panels before and after they talk.

Conformity and debate. Weng et al. (2025) show that LLM agents conform to the majority in groups. Multi-agent debate (Du et al., 2023) uses discussion to improve answers. Our split-evidence result shows the opposite risk: discussion can lose information that a simple average keeps.

Hidden profiles. In human groups, Stasser and Titus (1985) found that discussion favors shared information, so groups miss answers that only their pooled private information reveals. Our jigsaw arm reproduces this pattern in LLM panels.

Multi-agent risk. Hammond et al. (2025) name network effects and correlated failures as multi-agent risks. Our companion submission (DiMarco, 2026) measures herding in 538 games of AI social deduction and uses the same errors-only correction.

3. Methods

Cases. A deterministic generator writes 30 fictional village cases, each with 6 suspects, 3 traits per suspect and witness clues. In 20 consistent cases the clues identify one culprit. In 10 adversarial cases, two loud clues point at an innocent decoy, and one quiet clue rules the decoy out. A checker confirms that exactly one suspect fits all clues. A hard bank of 12 cases with 8 suspects rules out the decoy only through a negative sighting.

Panels. The main panel has six models from four families: two Qwen models (Qwen 3.5 397B, Qwen 3.8 27B), two Gemma models (Gemma 4 31B, Gemma 4 26B), GLM 5.2 and DeepSeek V4 Flash, all hosted by Adaption, at temperature 1.0. Each agent returns strict JSON probabilities over the suspects. A duplicate arm uses five Qwen 3.5 copies and one GLM 5.2.

Conditions. C1: one first read, no talk. C2: first read, one statement round, one reply round in which each agent sees the others' anonymized statements, then a final read.

C3: as C2, but each agent is told its first read is locked and will count. Devil's advocate: as C2, with agent 0 told to argue against the emerging favorite. Jigsaw: as C2, but each clue goes to 3 of the 6 agents by a fixed round-robin split. A second review found that this split gives only two distinct clue packets, each shared by three agents, and that a symbolic average of the two packets identifies the culprit in all 60 assignments. Each cell runs 30 cases × 2 seeds = 60 panels (57 for devil's advocate, 23 per arm on the hard bank).

Decision rules. *Pooled first reads*: the average of the six first reads, taken before any talk. *Discussed vote*: the average of the final reads. The panel is right when the top suspect of the average is the culprit.

Agreement. Our first draft correlated two agents' probability lists within one case. That measures how alike their answers look, not whether their errors depend on each other. The quantity that decides how much averaging helps is the correlation of errors across cases (Kish, 1965). For each pair of models in a panel, we take the error 1 − p(culprit) on each case and correlate it across the cases both answered, before and after talk. Effective number N_eff = n / (1 + (n − 1) r), with r clipped to [0, 1]. N_eff is an index, not a literal count: it assumes every pair shares one correlation, and it is computed on the defined pairs only. A model that is always certain and right has no error variance, so pairs that include it are undefined; 5 to 14 of 15 pairs are defined per arm. Intervals resample cases.

Statistics. We compare decision rules on the same panels with exact McNemar tests, and also by scenario, because the two seeds of a case are not independent. All analyses use the logged calls; nothing is typed by hand.

Data. The repository holds 11,402 logged records, including a stopped first sweep and a smoke test. The analysis uses 8,082 calls from four runs. 8–16% of estimates failed to parse, mostly from reasoning truncation, and are left out. Discussion statements were also often empty because the model returned no text: 43 of 360 in free discussion, 28 in locked framing, 67 of 342 with a devil's advocate and 69 of 360 (19%) in the jigsaw arm. A historical resume bug dropped 31 logged statements from the transcripts of 7 full-information panels (5 free discussion, 2 locked framing); it did not affect the jigsaw or devil's-advocate arms.

Pre-registration and deviations. Hypotheses H1–H4 were committed before the first model call (commit b01f872). We changed the panel size from 5 to 6, used 30 cases and 2 seeds, and did not run the placebo bank. The jigsaw and devil's-advocate arms were added later and are exploratory.

4. Results

4.1 Discussion makes errors line up across cases

We simulated panels of six agents that each solve a case alone, with independent noise. Within-case agreement on whole probability lists scores them as 1.02 effective

forecasters out of 6, the value we first reported as "collapse" (Figure 1, left). The second review built an exact counterexample of the same kind: six independently sampled estimates whose lists correlate perfectly, while their average still cuts random variance by a factor of 5.95 in 50,000 simulated panels.

On the right quantity, error correlation across cases, discussion does make errors line up (Figure 1, right; Table 1).

Table 1. Error correlation across cases, same model pairs, before and after talk. Interval: change, resampling cases.

Arm	Pairs	r before	r after	N_eff of 6	95% CI of change
Free discussion	5	0.42	0.77	1.95 → 1.24	−0.00 to 0.79
Locked framing	5	0.34	0.74	2.24 → 1.27	0.00 to 0.69
Devil's advocate	8	0.17	0.56	3.23 → 1.59	0.08 to 0.63
Jigsaw	14	0.22	0.46	2.84 → 1.81	−0.06 to 0.54

The direction is the same in all four arms. Two intervals exclude zero, and the pair counts are small. The second review's reanalysis, which keeps only panels where all six agents answered, points the same way: the gain from averaging (mean individual Brier minus panel Brier) falls after talk, by 0.014 [0.005, 0.029] in free discussion and by 0.081 [0.039, 0.114] in the jigsaw arm, on 4 and 6 cases.

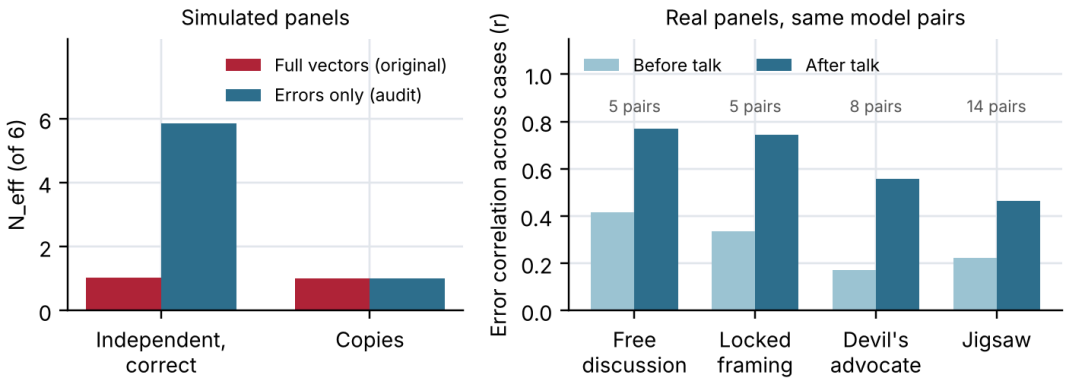


Figure 1. Left: effective number of forecasters for simulated panels of 6, on the original within-case measure and on errors only. Right: error correlation across cases before and after talk, on the same model pairs.

4.2 Pooling first reads beats the discussed vote when evidence is split

Table 2. Panel accuracy by decision rule (probability average), on the same panels. Left: each rule uses every valid read it has. Right: only agents with a valid read at both stages.

Arm	Panels	Pooled	Discussed	p	Pooled (matched)	Discussed (matched)	p
Full info, free discussion	60	60	59	1.00	59	59	1.00
Full info, locked framing	60	60	58	0.50	58	58	1.00
Full info, devil's advocate	57	57	57	1.00	57	57	1.00
Split evidence (jigsaw)	60	60	53	0.016	57	53	0.29
Hard bank, four families	23	23	23	1.00	23	23	1.00

With full evidence, every rule is near ceiling, and we cannot tell them apart. With split evidence, each agent's first read is right only 64% of the time, but the average of the six first reads names the culprit in all 60 panels. After discussion, individual agents improve to 80%, yet the panel drops to 53 of 60 (52 of 60 by plurality vote). All 7 changes go one way: the pooled reads were right and the discussed vote was wrong. The failures come from 5 of 30 cases (scenario-level sign test p = 0.06). In the 3 failed adversarial panels, the discussed vote chose the planted decoy each time.

Why this is suggestive, not established. The two rules do not use the same agents: on average only 4.4 of 6 agents have a valid read at both stages, because final reads fail to parse more often. On those matched agents, pooling is right in 57 panels and discussion in 53 (6 against 2 discordant, p = 0.29). Discussion also ran with gaps: 19% of jigsaw statements were empty, so some agents never shared their clues. Part of the drop may come from lost information, not from discussion itself.

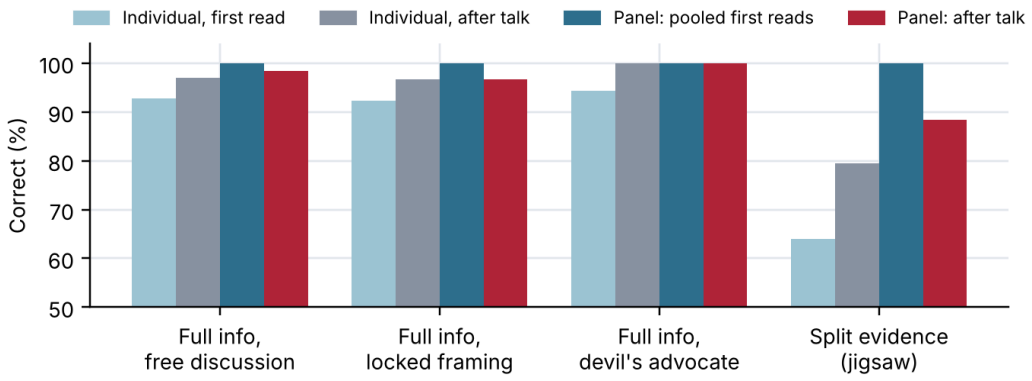


Figure 2. Individual accuracy and panel accuracy before and after talk. Discussion helps individuals in every arm, but in the jigsaw arm it costs the panel 7 of 60 decisions.

Reading. If it holds, this matches the hidden-profile pattern (Stasser and Titus, 1985). The information needed is spread across the panel. Averaging private reads combines it. Discussion pulls agents toward a story that fits the shared clues, and some panels settle on the wrong one.

4.3 Discussion lowers certainty

The share of agents giving 99% or more to one suspect fell after talk in every arm: from 74% to 51% in free discussion, 67% to 49% with locked framing, 76% to 29% with a devil's advocate, and 40% to 21% in the jigsaw arm. Discussion makes agents hedge, even when it makes them more accurate.

4.4 Retracted and unchanged claims

- **Deliberation collapses panels to one voice (H1): revised.** The original measure could not show this. On error correlation across cases, discussion does raise dependence (N_{eff} about 2–3 → 1.2–1.8), on few model pairs (Section 4.1).
- **Same-model premium (+0.082) (H3): retracted.** In the duplicate arm, Qwen 3.5 copies gave one-hot answers in 100% of first reads (GLM 5.2: 95%). Copies agree perfectly because each is certain and correct, so the comparison measures certainty. That it matched our companion study's +0.082 was a coincidence.
- **Locked first reads fail as a remedy (H4): revised.** Telling agents their first read is locked did not change their final reads. But using the locked first reads as the decision was never worse than the discussed vote, and with split evidence it was better on all available reads (not on matched agents).
- **Agreement-based collusion detection is unworkable: withdrawn.** It used the within-case measure, and its calibrated threshold (1.27) exceeds the largest possible correlation, so it could never flag a pair.
- **Danger band (H2): unchanged null.** Panels were too accurate for confident-but-wrong majorities to appear.

5. Discussion and Limitations

Implications. First, measure agent independence as error correlation across cases, not as similarity of answers within a case, or correct panels will look like herds. Second, a panel's private first reads are a resource that discussion can spend. Recording each agent's independent read before any talk, and pooling those reads, costs almost nothing. In our data it was never worse than discussion, and with split evidence it was better on all available reads, though not significantly on matched agents. Monitoring swarms often have split evidence: different monitors see different logs. Third, confident retractions matter: two of our first headline claims were measurement artifacts, and we found them only by testing the measure on cases with a known answer.

Limitations. The cases are synthetic logic puzzles at near-ceiling accuracy, so the full-evidence arms cannot separate the decision rules. The jigsaw result rests on 60 panels

from 30 cases, is suggestive at the scenario level ($p = 0.06$), is not significant on matched agents ($p = 0.29$), and ran with 19% empty discussion statements. The jigsaw arm was not pre-registered. Error correlation is undefined for models that are always certain and right, so Table 1 uses 5 to 14 of 15 model pairs and may favor less certain models. We used open-weight models on one host and one discussion format. Appendix A gives more detail.

Future work. Test pooled first reads against discussion on harder, open-ended tasks, with more cases per arm, and with evidence split the way real monitors see it.

6. Conclusion

Once we measured the right quantity, discussion did make LLM panels' errors line up across cases, though on few model pairs. When the evidence was split, the average of locked first reads beat the discussed vote, 60 to 53, but only 57 to 53 on matched agents, so the result is suggestive. Locking and pooling independent reads costs almost nothing; a pre-registered follow-up with complete transcripts should test whether it beats discussion.

7. Code and Data

Code, scenario banks and all 11,402 logged calls (prompts and outputs): github.com/vdimarco/apart-swarm-alignment. `python -m tests.test_analyze` runs the simulation check, `python -m src.across_case` gives Table 1, `python -m src.audit_stats` gives Sections 4.2 and 4.3, and `python -m src.audit` runs the second review's offline checks. All code, runs and analysis were done during the hackathon (November 7 to 8, 2026). No information hazard: the cases are fictional puzzles.

8. LLM Usage Statement

LLM coding agents wrote most of the code and ran the experiment. Two AI-assisted reviews audited the analysis separately; one had no access to the logs and worked from code and math. Claude (Anthropic) merged both, wrote the corrected statistics and drafted this report. Every number comes from scripts run on the published logs. The audit changed three headline claims, listed in Section 4.4 and Appendix B. The author reviewed the report and is responsible for its content. References were checked against their sources.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Limitations

- **Ceiling accuracy.** With full evidence, panels were right 97 to 100% of the time under every rule, so those arms cannot show a difference between pooling and discussion.
- **Sample size.** The jigsaw result is 7 discordant panels from 5 of 30 cases. The panel-level test ($p = 0.016$) treats the two seeds of a case as independent; the scenario-level test ($p = 0.06$) does not.
- **Exploratory arms.** The jigsaw and devil's-advocate arms were added after the pre-registration and should be read as exploratory.
- **Undefined agreement.** Error correlation needs error variance. Models that are always certain and right leave it undefined, so Table 1 uses 5 to 14 of 15 model pairs; the complete-panel check uses only 4 to 6 cases.
- **Data quality.** Final reads failed to parse more often than first reads, so the two decision rules can use different agents (Table 2, right). Discussion statements were empty in 8–20% of cases, and a resume bug dropped statements in 7 full-information panels.
- **Two packets.** The jigsaw split gives two distinct packets, not six. A balanced split with six packets is implemented but not run.
- **Setting.** Synthetic puzzles, open-weight models on one host, one discussion format, temperature 1.0, and 8–16% parse failures.

Dual-use and ethical considerations

The cases are fictional text puzzles. No people took part, no real systems were touched, and the adversarial cases mislead only the panels under study. The main dual-use risk is that knowing how discussion loses split evidence could help someone steer a panel of AI monitors toward a wrong answer. We judge this low, because the safeguard we describe, pooling locked first reads, is simple to adopt and is the main takeaway.

Appendix B. Audit: first-draft claims and what changed

Table 2. First-draft claims, the audit check, and the result.

First-draft claim	Check	Result
Discussion collapses panels to N_eff ≈ 1.0	Simulation and exact counterexample; error correlation across cases	Old measure cannot show it. Across cases, dependence rises in all four arms (Table 1). Revised.
Same-model premium +0.082 [0.074, 0.089]	Inspect what the duplicate arm compares; panel-level bootstrap	Copies are one-hot and correct; interval widens to [0.068, 0.095] even on the old measure. Retracted.
Locked first reads fail	Score the locked reads as the vote	Pooled locked reads: never worse; better with split evidence. Revised.
Jigsaw is a partial, lossy de-correlator	Compare decision rules on the same panels	The loss comes from discussion, not from the split: pooled reads 60/60, discussed vote 53/60.
Agreement-based collusion detection flags 66–97% of honest strangers	Measure used	Built on whole-vector agreement. Withdrawn.
Companion study is pre-existing work	Dates and authorship	It is a separate weekend submission on third-party data. Corrected.
Locked first reads carried into discussion	Fake-client probe of the original harness	They were never passed on, and resumed runs lost discussion text. Both fixed for new runs.
Pooled reads 60/60 vs discussed vote 53/60	Same agents at both stages; empty statements; resume bug	57 vs 53 on matched agents ($p = 0.29$); 19% empty jigsaw statements. Downgraded to suggestive.
Six model families	Model list	Six models from four families (two Qwen, two Gemma). Corrected.
Logs public	Repository contents	Logs were not in the repository. Now published.